

What a Windowed LLM Judge Cannot See: Provenance, Out-of-Horizon Self-Copy, and the Whole Document

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Abstract

Evaluating thousands of tokens of generated text with an LLM judge that reads one window at a time is common practice. We audit one such evaluation, a study of interventions on long, open-ended generation by base language models [12], in which Claude Opus 5 judged 96- to 160-token windows with 600 tokens of context, and report three problems, each large enough to change its conclusions. (1) *Provenance*: sentences the experimenter injected into the stream fell inside the judged windows and were scored as the model’s own; the judge could read them, but nothing told it who wrote them. (2) *The horizon*: given the same four injected sentences in rotation, the model replays its own earlier segments; at an injection period of 300 tokens, 65% of judged windows were more than half verbatim replay and the source of 62% lay outside the judge’s context, so replay was scored as surprise and as connection. (3) *Composition*: read as whole documents, no condition built an integrated text, and the condition with the best windows read as a sequence of restarts. Under a corrected protocol, the effect on surprise survives: at period 150 close to its first-reported size (+1.4 first reported, +1.2 on fresh windows), while the larger effect first reported at period 300 (+2.7) came from the injected text (+1.3 on generated text). The effect on connection was mostly replay (+2.1 first reported, +0.7 on fresh windows). The fresh-only estimates are stable across shingle lengths from 8 to 16 tokens and copy thresholds from 0.3 to 0.7. We present these as failure modes that long-form evaluations with windowed judges should test for.

1 Setting and terms

Ono Filho [12] asks where the novelty a base language model produces with no task comes from. Its unit is the *cell*: one premise (a single opening sentence) continued by a base model, mainly Qwen3-30B-A3B-Base (8-bit, MLX), for 4,500 tokens with the end-of-text token masked. An *arm* is a condition applied to every cell; ten premises are shared by all arms, and 24 arms were run. The terms used below:

Habituation: a repetition penalty over a moving window (the log-probability of every token seen in the last 512 generated tokens is lowered in proportion to its count), which removes literal loops.

Interruption: a short sentence injected into the model’s context as if the model had written it, introducing a new subject (for example “*Meanwhile, in a city with no name,*”). The *period* is the number of generated tokens between injections (150 to 900). The main arms rotate a fixed set of four such sentences.

Sham break: an injection without a new subject (a bare paragraph break, or a connective that asks for continuity), a control for injecting anything at all.

Reset: the context is cleared at each interruption, and the stream resumes from the premise plus the new subject.

Fresh window: a judged window that is not a verbatim replay of the stream’s own earlier text (Section 2.2).

Windows of the stream are judged by Claude Opus 5 on three scales from 0 to 10 (surprise given the earlier text, connection to it, coherence), five calls per window with the median as the score, with the 600 preceding tokens shown as context. The first version of the study, written under this protocol, reported that at period 150 the interruption raised surprise from 1.7 to 3.1 and connection from 1.1 to 3.2 over habituation alone; that its yield grew with the length of the thread it broke (5.3 to 5.9 in the windows after periods of 300 to 900 tokens, against 3.1 after 150); and that a period of about 300 tokens gave the best stream (surprise 4.35). Each of these numbers was partly an artefact of what the judge could and could not see.

2 Three things a windowed judge could not see

We order them by where the fault lies. The first is an error of the evaluation pipeline; the second and third are limits of judging a long stream by windows.

2.1 Provenance: an error of the pipeline

The first protocol cut 160-token windows right before each injection and on a uniform grid, so a window after an interruption contained the injected sentence. The judge could read that sentence. What it lacked was provenance: nothing in the window said that the experimenter, not the model, had written it, and a sentence that opens a new subject is surprising given the earlier text by construction. This is a segmentation error of the evaluation design, not a limit of the judge, and segmenting on provenance fixes it. Re-judging with windows of generated text only (96 tokens starting 32 tokens after the end of the injected sentence, none crossing an injection; a uniform grid in uninterrupted arms) kept the direction of every effect and removed the period story: on surprise, the post-interruption window scores 3.0 at period 150, 2.9 at 300, 2.9 at 600 and 3.2 at 900, and every paired difference between periods covers zero. Across 14 conditions, the condition means under the two protocols correlate at $\rho = +0.81$ (surprise) to $+0.93$ (connection); the sizes differ where an injected sentence sat in the window.

2.2 The horizon: replay the judge cannot recognize

A base model given the same four subject-change sentences in rotation learns the rotation: after “*That night she dreamed of something else entirely:*” it tends to write what it wrote the last time that sentence appeared, one rotation earlier. This is the self-reinforcing tendency of language models to repeat sentences already in their context [16], now cued by the experimenter. We flag a judged window as *copied* when at least half of its 12-token shingles occur earlier in the same stream, and record whether the source lies within the 600 tokens the judge sees (Table 1). Bare generation is 67% copied (literal loops, all in view, which the judge scores 0.0), habituation alone 27%, and the interruption arms 72% at period 150 and 65% at 300.

Where the source lies depends on the period. At 300, the rotation repeats every 1,200 tokens and the source of 62% of all windows lies outside the judge’s context: the judge cannot tell replay from new text and scores it as surprise and as connection. At 150, the rotation repeats every 600 tokens, so the source sits at the edge of the horizon and is partly in view; there the replay mainly

arm (Qwen3-30B-A3B-Base)	windows	copied	source beyond 600 tokens	fresh-only S / C / H
bare	60	67%	0%	1.77 / 1.38 / 5.55
habituation	60	27%	8%	1.76 / 1.32 / 5.01
habituation + interruption 150	60	72%	12%	2.95 / 2.03 / 6.78
habituation + interruption 300	60	65%	62%	3.02 / 2.03 / 6.52
habituation + interruption 600	60	38%	35%	2.84 / 2.04 / 6.33
habituation + interruption 900	50	12%	12%	3.12 / 2.25 / 6.75
sham break 300	60	32%	5%	2.16 / 1.41 / 5.03
reset + new subject 300	60	0%	0%	3.75 / 3.30 / 6.87

Table 1: Self-copy in the judged windows (the first generated-only window after each injection, or the grid windows of uninterrupted arms): share that is at least half verbatim reproduction of the stream’s own earlier text, share whose source lies beyond the 600 tokens shown to the judge, and cell means of surprise / connection / coherence on the fresh windows only (ten premises). At period 150 the location of the source depends on the detector (Appendix A).

period	windows judged	surprise	connection
150	first report (injected text inside windows)	+1.38	+2.07
150	generated text only, all windows	+1.42 [+0.92, +2.03]	+2.37 [+1.57, +3.28]
150	generated text only, fresh windows	+1.19 [+0.42, +2.09]	+0.71 [+0.36, +1.09]
300	first report (weighted stream estimate)	+2.65	+2.10
300	generated text only, all windows	+1.30 [+0.68, +1.92]	+1.08 [+0.47, +1.63]
300	generated text only, fresh windows	+1.25 [+0.71, +1.78]	+0.71 [+0.33, +1.20]

Table 2: The interruption’s effect over habituation alone, as first reported and under the corrected protocol. First-report values are differences of arm means in the first version of the study; corrected values are means of differences paired over the ten premises, with bootstrap 95% intervals over premises (every corrected contrast $p \leq 0.014$, exact sign-flip permutation). Surprise survives at period 150; at 300 the first report was the injected sentence. Connection was mostly replay at both periods.

inflates connection, because a verbatim reproduction of the earlier text is, literally, connected to it. At 600 and 900 copying falls to 38% and 12%, and the arm whose context is reset at each interruption has none. Removing the copied windows changes the interruption’s effect on surprise little and cuts its effect on connection by a third at period 300 and by more than two thirds at 150 (Table 2). We found this because a document-level judge (Section 2.3) remarked that the stream “repeats the same four-part block verbatim”.

2.3 Composition: local gains do not add up

Every score above is local. The judge also read the whole 4,500-token stream of each cell, injected sentences removed, and scored integration (parts taken up and joined later), development (something builds rather than restarts), coherence and surprise, once per document ($k = 3$). No condition exceeds 2.5 on any dimension. The interrupted stream over the preserved context, whose windows are the best, reads *below* the uninterrupted habituated one (surprise 0.1 against 1.4, $p = 0.002$; development 0.3 against 1.1): a sequence of restarts. The reset arm reads best (2.0 / 1.6 / 2.2 / 2.5), because each of its parts is a clean piece that returns to the premise; this replicated on ten new premises.

3 The protocol that corrects them

1. **Windows segmented on provenance:** generated text only, starting a margin after any injected text and never crossing an injection. The context shown to the judge may contain

injected text; the graded region never does.

2. **A self-copy flag and fresh-only estimates**, reported next to the all-windows ones, with the detector’s sensitivity checked (Appendix A). Across shingle lengths of 8, 12 and 16 tokens and thresholds of 0.3, 0.5 and 0.7, the fresh-only effect on surprise stays between +1.10 and +1.29 (every $p \leq 0.010$) and on connection between +0.68 and +1.18 (every $p \leq 0.023$), always well below the all-windows connection effect at period 150 (+2.37); copied shares move by at most ten points. The one detector-dependent quantity is where the source lies at period 150 (between 5% and 70% of windows beyond the horizon), because there it sits at the horizon’s edge; at 300 it is 60% to 62% under every setting. The flag detects verbatim replay only. A design rule follows: never inject the same sentence twice, or reset the context.
3. **A document-level reading** of the whole stream, with injected text removed, as the check that local gains compose.
4. **The stream, not the window, as the unit of inference**: cell means, bootstrap intervals over cells, exact sign-flip permutation tests on paired cells; window-level statistics only as descriptive.
5. **A measured judge**: the median of five calls; intra-window spread 0.45 to 0.70; a pipeline test–retest (two arms that turned out byte-identical in all ten cells were judged independently: medians of five agree exactly in 91% of windows on surprise, and cell means differ by -0.01 [$-0.09, +0.07$]); and a second judge family, Kimi K2.6, which agrees at $\rho = 0.85$ / 0.77 / 0.71 (surprise / connection / coherence) on a stratified sample of 140 first-protocol windows and at 0.75 / 0.80 / 0.65 on a random sample of 160 generated-only windows, reproducing the ordering of conditions in both.
6. **Human raters, with their reach stated**. Three raters recruited on a freelance platform, blind to hypotheses and conditions, rated 63 first-protocol windows (nine per condition, stratified by the judge’s surprise score) on surprise and coherence. Their consensus agrees with the judge at $\rho = 0.58$ on surprise ($p = 7 \times 10^{-7}$; single raters 0.40 to 0.50; inter-rater Krippendorff $\alpha = 0.30$) and reproduces the ordering of conditions; on coherence they rank the interrupted arms below the habituated one, which the judge does not. These data support the ordering of conditions on surprise and a moderate agreement with the judge. They do not validate the corrected effect sizes, connection, or the document level, which rest on LLM judges; a second round on fresh generated-only windows was designed but not run.
7. **A pre-registered confirmatory battery** on ten new premises and a new random seed, analyzed under the protocol fixed in advance: the primary contrast confirmed (+1.48 [$+0.70, +2.07$], one-sided $p = 0.003$; on fresh windows +0.82 [$+0.03, +1.44$], $p = 0.035$), and the expectation that a kept context carries connection refuted.

4 What survives, and what generalizes

The interruption is real. On fresh generated text it raises judged surprise over habituation alone by +1.2 to +1.3 on the original premises and by +0.8 on the confirmatory ones. At period 150 this is close to the size first reported; the larger effect first reported at period 300, and its growth with the period, came from the injected sentence. Its effect on connection was mostly replay. In direction the effect replicates, on generated-only windows, on three base models from two families, a post-trained model, an unquantized model and a second genre [12]. A bare paragraph break adds nothing detectable, and a connective that asks for continuity hurts. What did not survive were the claims that depended on what the judge could not see.

What we think generalizes is narrower than a law and more useful than an anecdote. When an output does not fit the judge, the evaluation pipeline decides what the judge sees, and three

of its choices can create or erase an effect. Long-form evaluations with windowed judges should test whether judged windows contain text the model did not write; whether the model replays material whose source lies outside the judge’s context; and whether local scores compose into the quality of the whole. We showed each in one unusual setting, unconstrained 4,500-token base-model continuations with periodic injections; how often they matter in other long-form evaluations is an open empirical question.

5 Related work

LLM judges and their biases. Strong LLMs are now standard evaluators [4, 10, 17], and their biases are well documented: sensitivity to the order of candidates [14] and a preference for their own generations [13] and for longer outputs [6]. Recent work asks whether a judge tracks the property it is meant to measure: Chen et al. [3] separate a judge’s invariance to construct-preserving edits from its sensitivity to construct-changing ones and find that high reliability can coexist with low sensitivity, and Dev et al. [5] stress-test judge reliability across benchmarks. These studies examine how a judge scores an output it sees whole. Our case concerns a different axis, the unit of judgment: when an output is longer than the judge’s context, the pipeline decides what the judge sees, and that decision can create or erase an effect.

Evaluating long outputs. Long outputs are commonly evaluated by decomposition: into atomic facts [11], into units for human annotation [8], or into chunks whose summaries are merged, where coherence across chunks is itself a known difficulty [2]. A longer judge context is not a full remedy, since models use long contexts unevenly [9]. Local scores that do not compose into document quality are a judged-quality instance of the same decomposition problem.

Repetition in generation. Degenerate repetition under likelihood-maximizing decoding is long known [7, 15], and sentence-level repetition reinforces itself: the more often a sentence has appeared in the context, the more likely it is to be repeated [16]. The self-copy we report is that tendency meeting a fixed rotation of injected cues, with the copy’s source outside the judge’s view. For creative text, Chakrabarty et al. [1] found that no LLM used as an assessor correlated positively with expert judgments, one more reason to measure the judge rather than assume it.

6 Limitations

The evidence comes from one study: ten premises (plus ten confirmatory ones), one unusual generation setup, one main judge family with a second family on samples. Human validation is thin, as item 6 of the protocol states. The self-copy flag detects verbatim replay only; paraphrased self-copy would pass as fresh and, if present, would inflate the fresh-only estimate of connection. The document-level reading uses the same judge family, with three calls per document.

Acknowledgments and tooling

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Code and data

The code of the audited study is in the repository at <https://github.com/RobertoOno/interrupting-the-loop> as of 10 September 2026; the complete code, including the self-copy sensitivity script and its appendix, the run data and the LLM judgments behind every number in this paper are in the evidence deposit of the audited study ([doi:10.5281/zenodo.23163299](https://doi.org/10.5281/zenodo.23163299)). Individual human ratings are not released, to protect rater anonymity; only aggregate statistics are reported.

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A Sensitivity of the self-copy detector

A window is copied when a share of at least t of its K -token shingles occurs earlier in the same stream, and its source is beyond the horizon when the share seen within the 600 tokens shown to the judge is below t . Table 3 recomputes the copied shares and the fresh-only contrasts for every combination; the primary detector is $K = 12$, $t = 0.5$. The habituation arm’s copied share stays between 22% and 28%, and the reset arm has no copied window under any setting.

K	t	interruption 150		interruption 300		Δ surprise		Δ connection	
		copied	beyond	copied	beyond	150	300	150	300
8	0.3	77%	5%	65%	62%	+1.25	+1.22	+0.69	+0.68
8	0.5	73%	10%	65%	62%	+1.18	+1.22	+0.70	+0.68
8	0.7	70%	70%	65%	62%	+1.11	+1.28	+0.87	+0.72
12	0.3	75%	5%	65%	62%	+1.23	+1.22	+0.70	+0.68
12	0.5	72%	12%	65%	62%	+1.19	+1.25	+0.71	+0.71
12	0.7	68%	68%	63%	60%	+1.10	+1.26	+0.90	+0.71
16	0.3	72%	8%	65%	62%	+1.15	+1.22	+0.68	+0.68
16	0.5	72%	52%	65%	62%	+1.21	+1.28	+0.72	+0.72
16	0.7	67%	67%	63%	60%	+1.23	+1.29	+1.18	+0.73

Table 3: Copied share, share with the source beyond the judge’s horizon, and fresh-only contrasts of the interruption over habituation (paired over ten premises; every surprise contrast $p \leq 0.010$, every connection contrast $p \leq 0.023$, exact sign-flip permutation) for each shingle length K and threshold t . For reference, on all windows the effects are +1.42 and +1.30 on surprise and +2.37 and +1.08 on connection (periods 150 and 300). At period 150 the four-sentence rotation repeats every 600 tokens, so the source sits at the edge of the judge’s context and its classification depends on the threshold; at period 300 it does not.